CO4-AT2

NAME : LOGESHWARI S **REG NO : 192565040**

COURSE CODE:CSA0303

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QUESTIONS

Q1
Bubbled Sorting
25 marks

Q2
Insertion Soring
25 marks

Q3
Quick Sorting
25 marks

Q4
Merger Sort
25 marks

4/4 answered

Q1 Bubbled Sorting

25 marks

Write a C code for performing Bubbled Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```c
#include <stdio.h>

int main() {
    int A[] = {8, 2, 3, 5, 7};
    int n = 5, i, j, temp;

    for (i = 0; i < n - 1; i++)
        for (j = 0; j < n - i - 1; j++)
            if (A[j] > A[j + 1]) {
                temp = A[j];
                A[j] = A[j + 1];
                A[j + 1] = temp;
            }

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
        printf("%d ", A[i]);

    return 0;
}
```

Input

```
Enter 5 elements: 8 2 3 5 7
```

Output

```
Sorted array: 2 3 5 7 8
```

Submitted

Q2 .

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QUESTIONS

- Q1 Bubbled Sorting 25 marks
- Q2 Insertion Soring 25 marks
- Q3 Quick Sorting 25 marks
- Q4 Merger Sort 25 marks

4/4 answered

Q2 **Insertion Soring** 25 marks

Write a C code for performing Insertion Sort - n=5 A[] = {8 2 3 5 7}.

Your Code

```c
#include <stdio.h>

int main() {
    int A[5], i, j, key;

    printf("Enter 5 elements: ");
    for(i = 0; i < 5; i++)
        scanf("%d", &A[i]);

    for(i = 1; i < 5; i++) {
        key = A[i];
        j = i - 1;

        while(j >= 0 && A[j] > key) {
            A[j + 1] = A[j];
            j--;
        }
        A[j + 1] = key;
    }

    printf("Sorted array: ");
    for(i = 0; i < 5; i++)
        printf("%d ", A[i]);

    return 0;
}
```

Input

```
Enter 5 elements: 8 2 3 5 7
```

Output

```
Enter 5 elements: Sorted array: -664795216 0 0 1 69
```

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QUESTIONS

- Q1 — Bubbled Sorting — 25 marks
- Q2 — Insertion Soring — 25 marks
- Q3 — Quick Sorting — 25 marks
- Q4 — Merger Sort — 25 marks

4/4 answered

Q3 Quick Sorting

25 marks

Write a C code for performing Quick Sort - n=7 A[] = {8, 2, 3, 5, 7,1,6}.

Your Code

```c
#include <stdio.h>

void quickSort(int a[], int low, int high) {
    int i = low, j = high, pivot = a[(low + high) / 2], temp;

    while (i <= j) {
        while (a[i] < pivot) i++;
        while (a[j] > pivot) j--;

        if (i <= j) {
            temp = a[i];
            a[i] = a[j];
            a[j] = temp;
            i++;
            j--;
        }
    }

    if (low < j) quickSort(a, low, j);
    if (i < high) quickSort(a, i, high);
}

int main() {
    int A[7], i;

    printf("Enter 7 elements: ");
    for(i = 0; i < 7; i++)
        scanf("%d", &A[i]);

    quickSort(A, 0, 6);

    printf("Sorted array: ");
    for(i = 0; i < 7; i++)
        printf("%d ", A[i]);

    return 0;
}
```

Input

```
Enter 7 elements: 8 2 3 5 7 1 6
```

Output

```
Enter 7 elements: Sorted array: 0 0 1 16 69 694 296421568
```

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QUESTIONS

- Q1 Bubbled Sorting 25 marks
- Q2 Insertion Soring 25 marks
- Q3 Quick Sorting 25 marks
- Q4 Merger Sort 25 marks

4/4 answered

Q4 Merger Sort

25 marks

Write a C code for performing Merger Sort - n=8 A[] = {8 2 3 5 7 6 4 1}.

Your Code

```c
#include <stdio.h>

void merge(int a[], int l, int m, int r) {
    int i = l, j = m + 1, k = 0, b[8];

    while (i <= m && j <= r)
        b[k++] = (a[i] < a[j]) ? a[i++] : a[j++];

    while (i <= m) b[k++] = a[i++];
    while (j <= r) b[k++] = a[j++];

    for (i = l, k = 0; i <= r; i++)
        a[i] = b[k++];
}

void mergeSort(int a[], int l, int r) {
    if (l < r) {
        int m = (l + r) / 2;
        mergeSort(a, l, m);
        mergeSort(a, m + 1, r);
        merge(a, l, m, r);
    }
}

int main() {
    int A[8], i;

    printf("Enter 8 elements: ");
    for (i = 0; i < 8; i++)
        scanf("%d", &A[i]);

    mergeSort(A, 0, 7);

    printf("Sorted array: ");
    for (i = 0; i < 8; i++)
        printf("%d ", A[i]);

    return 0;
}
```

Input

```
Enter 8 elements: 8 2 3 5 7 6 4 1
```

Output

```
Enter 8 elements: Sorted array: 0 0 0 1 16 69 729 105991360
```

Submitted